

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA02

COURSE NAME: C Programming

COURSE OUTCOME COVERED

CO4: *Implement structures, unions, and nested structures for modeling and processing real-world data. (BL4,BL5)*

Assessment Tool Weightage: 40 %

QUESTIONS

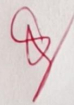
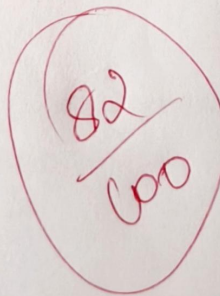
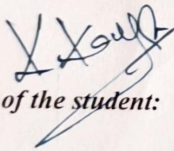
Total Marks: 100

Saveetha Compiler assessment (Level 2& Viva Voce)

Code of Conduct:

I K Kaung 192625283 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Subarray:

```
#include <stdio.h>
int main()
```

```
{
```

```
    int n, sum, i, j;
```

```
    scanf("%d", &n);
```

```
    int a[n];
```

```
    for (i=0; i<n; i++)
```

```
        scanf("%d", &a[i]);
```

```
    scanf("%d", &sum);
```

```
    for (i=0; i<n; i++)
```

```
    {
        int j=0;
```

```
        for (j=i; j<n; j++) {
```

```
            s+=a[j];
```

```
            if (s==sum) {
```

```
                printf ("Subarray with given sum: [%i]",
```

```
                for (int k=i; k<=j; k++)
```

```
                    printf ("%d", a[k]);
```

```
                printf ("");
```

```
                return 0;
```

```
            }
```

```
        }
```

```
        printf ("No subarray found");
```

```
        return 0;
```

```
    }
```

```
}
```

```
input
```

```
5
```

```
1 4 20 3 10
```

```
33
```

```
output
```

```
20 3 10
```


2. Concatenate

```
#include <stdio.h>
#include <string.h>

int main()
{
    char s1[20] = "Hello";
    char s2[10] = "World";
    strcat(s1, s2);
    printf("%s", s1);
    return 0;
}
```

Input

Hello

World

Output

HelloWorld

} in question

3. Sum of diagonal elements:

```
#include <stdio.h>

int main()
{
    int a[10][10], r, c, i, j, sum = 0;
    scanf("%d %d", &r, &c);
    for (i = 0; i < r; i++)
        for (j = 0; j < c; j++)
            scanf("%d", &a[i][j]);
    for (i = 0; i < r; i++)
        sum += a[i][i];
    printf("%d", sum);
    return 0;
}
```

input: 3
 3
 1 2 3
 4 5 6
 output 7 8 9
 15

4. Student

```
#include <stdio.h>
```

```
struct student
```

```
{ char n[10];
```

```
  int i, m[3];
```

```
};
```

```
int main()
```

```
{ struct student s;
```

```
  scanf ("%s %d %d %d", s.n, &s.i, &s, m[0], &s, m[0], &s, m[0], &s, m[2]);
```

```
  printf ("%s\n %d\n %d\n %d\n", s.n, s.i, s.m[0], s.m[2], s.m[2]);
```

```
}
```

input:

david

102

85

90

88

output:

david

102

85

90

88

~~Q 2/06/26~~

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Logic Development	25%	Develops correct, efficient, and well-structured logic for the problem	Logic is mostly correct with minor gaps	Logic is partially correct; needs improvement	Incorrect or no logical approach
Syntax & Semantics	20%	Writes error-free code with correct syntax and semantics	Minor syntax/semantic errors that do not affect execution significantly	Multiple errors; requires corrections	Frequent errors; code does not compile/execute
Time & Space Complexity	20%	Demonstrates optimal time and space complexity with proper justification	Reasonably efficient with minor scope for improvement	Limited consideration of complexity	No consideration of time or space complexity
Test Case Validation	20%	Designs and validates comprehensive test cases including edge cases	Tests cover most scenarios with minor gaps	Limited test cases; misses important cases	No or incorrect test case validation
Viva Voce	15%	Explains logic, code, and concepts clearly with confidence and depth	Explains concepts with minor gaps	Limited understanding; needs prompting	Unable to explain concepts